BARANGAY HEALTH STATUS MONITORING SYSTEM

A Capstone Project Presented to

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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Katlyne Biendima

Christian Angel Parado

Ryan Justine Bisnan

Dave Lester Bisnan

Mark Bolanos

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CHAPTER I

Introduction

Background of the Study

In today's rapidly evolving world, information technology plays a crucial role in improving efficiency, accessibility, and decision-making across various sectors, particularly in healthcare. The integration of digital systems into healthcare services has significantly transformed the way patient information is collected, stored, and analyzed. Globally, health information systems have been shown to enhance data accuracy, accessibility, and efficiency by enabling real-time monitoring

and informed decision-making. According to the World Health Organization (2022), digital health technologies contribute to improved healthcare outcomes through better data management and accessibility.

In recent years, the demand for digital health solutions has increased, especially in local communities where access to efficient healthcare services remains limited. Modern systems now utilize centralized databases and data analytics to support healthcare providers in monitoring patient conditions and identifying health risks. These developments highlight the importance of transitioning from traditional manual systems to automated and technology-driven solutions.

In the Philippines, the Department of Health (DOH) continues to promote digital health initiatives to improve healthcare delivery at the barangay level. Barangays serve as the frontline providers of essential health services such as immunization, maternal care, and disease prevention. However, many barangays still rely on paper-based systems in managing health records. As emphasized by Altura et al. (2023), this reliance on manual processes limits efficiency and highlights the need for a centralized digital platform to improve healthcare service delivery.

Manual record-keeping systems present several challenges, including data redundancy, loss of records, and delays in retrieving important information. Batoon et al. (2022) explained that these inefficiencies negatively affect the productivity of health workers and hinder the delivery of timely and accurate healthcare services. As a result, healthcare providers may struggle to monitor patient conditions effectively and respond to health concerns promptly.

To address these challenges, digital health systems have been developed to enhance healthcare accessibility and efficiency. Garcia et al. (2021) demonstrated that mobile and web-based applications can improve patient monitoring, medical record management, and communication between healthcare providers and residents. Their study showed that such systems can bridge the gap between communities and health centers, particularly in rural areas where access to healthcare services is limited.

Despite these advancements, several barangays—including Barangay Palaquio, Tabiog, and Babangcag—continue to rely on traditional methods in managing health information. Records are often stored in logbooks and physical files, making them prone to errors, duplication, and damage. Retrieving information becomes time-consuming, especially when handling large volumes

of data, which affects the efficiency of healthcare service delivery.

Another challenge faced by barangay health centers is the limitation in manpower and resources. With a limited number of health workers, continuous monitoring of residents becomes difficult, particularly for individuals who require regular health assessment. This limitation can lead to delayed detection of illnesses and reduced quality of care.

Furthermore, the absence of a centralized system results in fragmented health records and difficulty in generating accurate reports. Without real-time data, healthcare providers are unable to effectively track health trends or identify high-risk individuals within the community. Communication also remains a concern, as traditional methods such as face-to-face interaction and printed materials may not be sufficient for timely information dissemination. Sumaylo (2013) noted that such communication approaches can lead to delays and gaps in delivering important health information to residents.

Given these challenges, there is a clear need for a comprehensive and integrated system that can improve the management of health data in barangays. Therefore, this study proposes the development of a Barangay Health Monitoring System,

which will serve as a centralized platform for collecting, storing, and analyzing health information. The system aims to enhance efficiency, accuracy, and accessibility of health data while enabling health workers to monitor records, identify high-risk individuals, and generate real-time reports to support better healthcare service delivery.

Conceptual Framework

This study is anchored on the **Input–Process–Output (IPO) Model**, which illustrates how the proposed Barangay Health Monitoring System transforms data and user requirements into meaningful outputs that support efficient health information management and decision-making.

The **Input** component includes the necessary data and resources required for the development and operation of the system. This consists of user needs and system requirements gathered through interviews, surveys, and document analysis. It also includes residents' health data such as personal information, vital signs, medical history, vaccination records, and other relevant health information. Additionally, system resources such as hardware, software, internet connectivity, and users (barangay health workers, midwives, and administrators) are included.

The **Process** component represents the procedures and system operations that transform the inputs into useful outputs. This includes data encoding and validation, storage in a centralized database, and data processing and analysis. It also involves health status monitoring, report generation, and data visualization. Furthermore, the process includes system development stages such as planning, design, prototyping, testing, and refinement, as well as user interaction such as accessing, updating, and monitoring data within the system.

The **Output** component consists of the results produced after processing the input data. These include organized and updated health records, generated reports, identified health trends or risks, and improved access to accurate and timely information. These outputs support better decision-making and enhance healthcare service delivery in the barangay.

This conceptual framework demonstrates how the proposed system ensures efficient, accurate, and reliable health data management through a structured process, ultimately improving the quality of healthcare services.

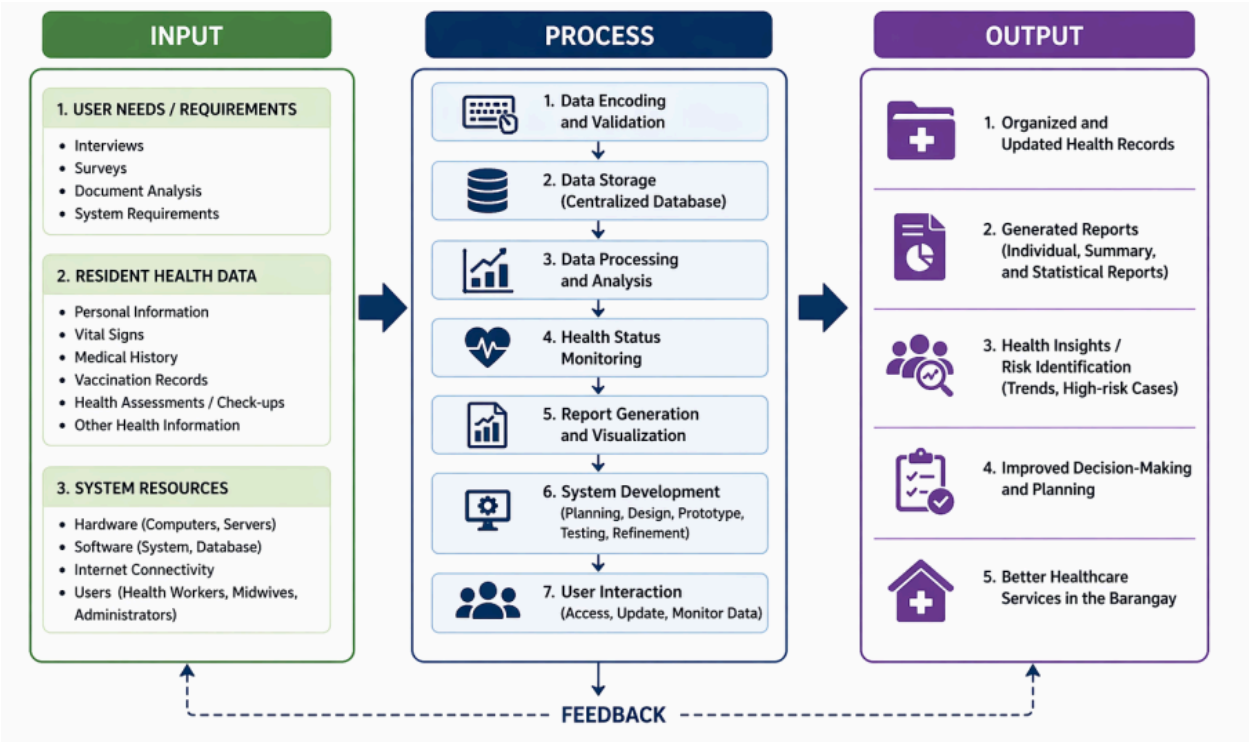


FIGURE1:IPO MODEL FOR BARANGAY HEALTH STATUS MONITORING SYSTEM

Statement of the Problem

The barangay currently faces challenges in managing and monitoring health data due to the lack of a computerized and centralized system. Manual processes lead to inefficiencies such as data redundancy, difficulty in retrieving records, and delays in reporting health information.

Specifically, this study seeks to answer the following questions:

1. What challenges are encountered in the current manual health monitoring system in the barangay?

2. What improvements can be made in the recording, storage, and management of health data through the proposed system?
3. What is the level of effectiveness, efficiency, and user-friendliness of the developed system as evaluated by the users?

Objectives of the Study

To develop a **Barangay Health Monitoring System** that will centralize and improve the management of health data of barangay residents.

Specific Objectives

1. To identify and analyze the challenges encountered in the current manual health monitoring system in the barangay.
2. To design and develop a system that improves the recording, storage, and management of residents' health data.
3. To determine the effectiveness, efficiency, and user-friendliness of the developed system based on user evaluation.

Scope and Delimitation

This study focuses on the design and development of a web-based Barangay Health Monitoring System intended to manage and monitor health data within selected barangays. The system

covers functionalities such as resident registration, health record management, vaccination tracking, disease monitoring, and automated report generation.

The primary users of the system include barangay health workers (BHWs), midwives, and barangay officials who are directly involved in healthcare service delivery. The system is designed to be used within the barangay health center and supports the monitoring of priority populations based on existing health programs.

However, the study is limited to the implementation within selected barangays only and does not include integration with national health databases such as DOH systems. The system is web-based and does not include mobile application deployment or SMS notification features in its current version. Additionally, advanced analytics such as predictive modeling and AI-based diagnosis are beyond the scope of this study.

Significance of the Study

This study will benefit the following:

Barangay Health Workers (BHWs**)** – This study will improve their efficiency in monitoring and managing health records of children, pregnant women, and adults by reducing manual workload and minimizing errors.

Midwives – This study will provide accurate and accessible patient data, which will help improve healthcare delivery, particularly in maternal care and adult health monitoring, and support timely interventions.

Barangay Officials – This study will support decision-making through reliable reports and analytics, enabling better planning and implementation of health programs for priority groups.

Residents (Children, Pregnant Women, and Adults) – This study will ensure better health monitoring, promote early detection of diseases, and improve access to healthcare services and information.

Future Researchers – This study will serve as a reference for future research and development of similar health information systems focused on specific population groups.

Sustainable Development Goals (SDGs)

This study supports the Sustainable Development Goals of the United Nations:

SDG 3: Good Health and Well-being – This study promotes better healthcare services by improving monitoring and disease tracking among children, pregnant women, and adults.

SDG 9: Industry, Innovation, and Infrastructure – This study contributes to innovation by developing a web-based system that enhances digital healthcare infrastructure.

SDG 11: Sustainable Cities and Communities – This study helps build healthier communities through improved local health management systems.

Definition of Terms

Barangay Health Information System – In this study, it refers to the web-based system developed to collect, store, manage, and analyze residents' health data in the barangay.

Centralized Database – In this study, it refers to the system's main storage where all residents' health records are securely kept and can be accessed by authorized users.

Dashboard – In this study, it refers to the system interface that displays summarized health data, reports, and key indicators for monitoring purposes.

Data Analytics – In this study, it refers to the system's ability to analyze stored health data to identify trends, patterns, and high-risk individuals.

Data Encoding – In this study, it refers to the process performed by health workers in entering residents' information and health records into the system.

Data Management – In this study, it refers to the system's function of organizing, storing, updating, and maintaining health data efficiently.

Data Privacy – In this study, it refers to the protection of residents' personal and health information by restricting access to authorized users only.

Data Security – In this study, it refers to the measures implemented in the system to prevent unauthorized access, data loss, or data breaches.

Disease Monitoring – In this study, it refers to the system's function of tracking and recording illnesses affecting residents over time.

Efficiency – In this study, it refers to the system's ability to reduce time and effort in recording, retrieving, and managing health data.

Health Indicators – In this study, it refers to measurable data such as blood pressure, temperature, and vaccination status recorded in the system to assess residents' health conditions.

Health Monitoring – In this study, it refers to the continuous process of recording and tracking the health status of children, pregnant women, and adults using the system.

Health Record – In this study, it refers to the digital file containing a resident's personal information, medical history, and recorded health data stored in the system.

Prototype Model – In this study, it refers to the development approach used in creating the system, where an initial version is built, tested, and improved based on user feedback.

Report Generation – In this study, it refers to the system's ability to automatically produce organized reports based on stored health data.

Resident – In this study, it refers specifically to children, pregnant women, and adults living in the barangay whose health information is recorded and monitored in the system.

Reliability – In this study, it refers to the system's ability to function consistently and accurately without errors during use.

Risk Level – In this study, it refers to the classification assigned by the system to residents based on their health condition and recorded data.

System Evaluation – In this study, it refers to the process of assessing the system's performance based on usability, efficiency, and reliability through user feedback.

System User – In this study, it refers to authorized individuals such as barangay health workers, nurses, midwives, and officials who use the system.

User Interface (UI) – In this study, it refers to the visual design of the system that allows users to interact with its features and functions.

Vaccination – In this study, it refers to the recorded administration of vaccines to residents as stored and tracked in the system.

Web-Based System – In this study, it refers to the system that can be accessed through a web browser within the barangay health center.

CHAPTER II

METHODOLOGY

This chapter presents the research methodology used in the study. It describes the research design, data gathering techniques, sampling procedure, system development process, and tools used in developing and evaluating the Barangay Health Monitoring System.

Research Design

The study utilized a **descriptive-developmental research design**. It is descriptive because it aims to identify and analyze the current challenges encountered in managing health

data in the barangay. It is developmental because it involves the design and development of a **Barangay Health Monitoring System** tailored to improve health data management.

The developmental aspect follows a system development approach to ensure a structured and systematic process in building the system—from requirement analysis up to testing and implementation.

System Development Approach

The system was developed using the Prototype Model, which allows iterative development through continuous user feedback. The process begins with **requirements gathering**, where user needs are identified through interviews and document analysis. This is followed by the **design phase**, where system architecture, database structure, and interface layouts are created.

In the **prototyping phase**, an initial version of the system is developed and presented to users for evaluation. The **testing phase** involves assessing system functionality, usability, and performance. Based on feedback, the system undergoes **refinement**, where improvements and corrections are applied. Finally, the **deployment and evaluation phase** ensures that the system meets user requirements and operates effectively within the barangay setting.

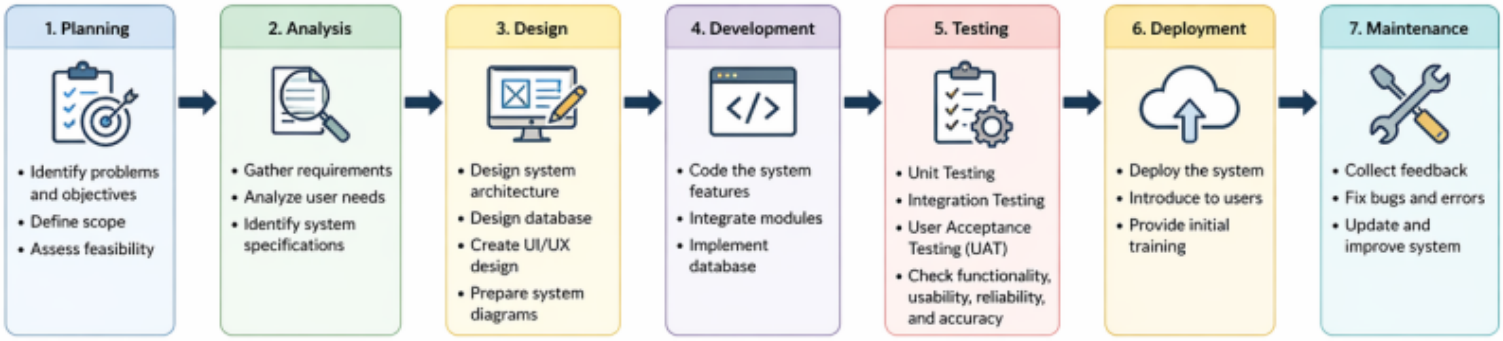


Figure 2. Prototype Development Cycle

PHASES IN SDLC

Planning Phase-In this phase, the researchers identified the existing problems in the barangay's health monitoring process, such as difficulty in managing records, data redundancy, and delays in retrieving information. The objectives, scope, and feasibility of the proposed Barangay Health Monitoring System were also defined to ensure that the project is achievable and relevant.

Analysis Phase-During this phase, the researchers gathered and analyzed system requirements through interviews, surveys, and document analysis. The needs of the users, particularly barangay health workers and midwives, were identified to ensure that the system addresses real-world challenges and requirements.

Design Phase-In the design phase, the overall structure of the system was planned. This includes designing the user interface for easy navigation, creating the database structure for storing health data, and developing system diagrams such as Use Case Diagrams and Data Flow Diagrams (DFD). This phase serves as the blueprint for system development.

Development Phase-In this phase, the actual system was developed based on the approved design. The researchers coded the system features, integrated different modules such as data input, processing, and reporting, and implemented the database. This phase transforms the design into a working system.

Testing Phase-The developed system was tested to ensure that it functions correctly and meets user requirements. Different types of testing were conducted, including unit testing, integration testing, and user acceptance testing (UAT). The system was evaluated based on functionality, usability, reliability, and accuracy.

Deployment Phase-After successful testing, the system was deployed and introduced to the intended users. Barangay health workers and midwives began using the system for managing health data in the barangay.

Maintenance Phase-In this final phase, the system was continuously monitored and improved after deployment. Feedback from users was collected to identify issues and areas for improvement. Necessary updates, bug fixes, and enhancements were applied to ensure the system remains efficient and reliable.

System Development Methodology (Prototype Model)

This study follows the **Prototype Model** in developing the Barangay Health Monitoring System, which focuses on iterative development and continuous user feedback.

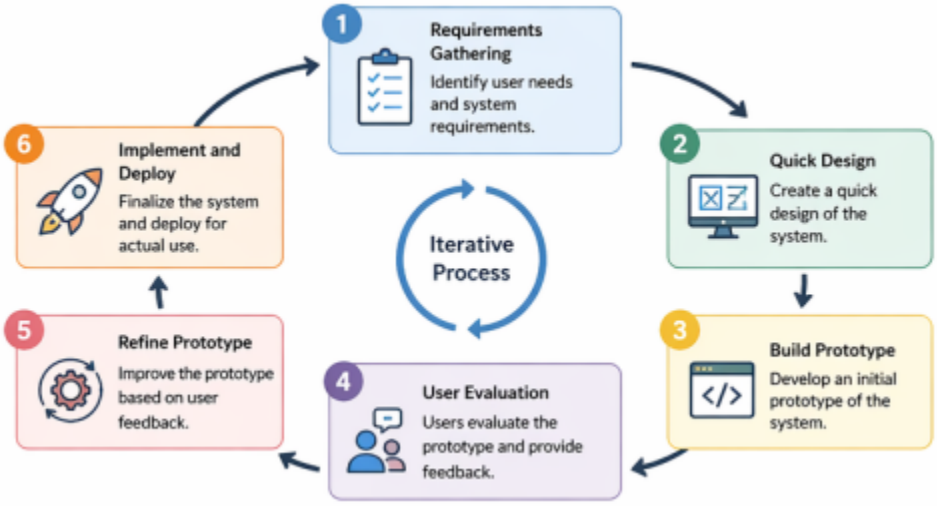


Figure 3. Prototype Development Cycle

Requirements Gathering – The researchers identified user needs such as efficient data recording, easy access to health records, and report generation.

Design – The system interface and structure were designed to be user-friendly for barangay health workers.

Prototyping – An initial version of the system was developed for user evaluation.

Testing – The system was tested to check functionality, usability, and performance.

Refinement – Improvements were made based on user feedback.

Deployment and Evaluation – The system was implemented and evaluated to ensure it meets user requirements.

Development Tools and Technologies

The development of the Barangay Health Monitoring System utilized various tools and technologies to ensure efficiency, reliability, and functionality.

The system was developed using **HTML, CSS, and JavaScript** for the front-end interface, providing a user-friendly and interactive design. For the back-end, **PHP** was used to handle

system logic, data processing, and communication between components.

A **MySQL database** was utilized to store and manage health-related data, including resident information, health records, and reports. The system was developed using an Integrated Development Environment (IDE) such as **Visual Studio Code**, which supports efficient coding and debugging.

Additionally, **XAMPP (Apache Server)** was used as the local server environment for testing and deployment. Tools such as **Figma** were used for user interface design, while **Git** was utilized for version control and system management.

Sampling Techniques

The study used a **purposive sampling technique**, where respondents were selected based on their roles and involvement in barangay health services. A total of **10 respondents** participated in the evaluation of the system.

The respondents of the study include **barangay health workers (BHWs) and midwives**, as they are the primary users of the system. These respondents were selected because they are directly involved in managing and monitoring health data in the barangay.

Data Gathering Procedure

The researchers followed these steps in gathering data to conducted interviews with barangay health workers and midwives to identify current challenges. Will gather user requirements and feedback by distributing survey questionnaire. Will developed the prototype system based on collected data. Will conducted system testing and evaluation with respondents. After the collected and analyzed feedback for system improvement.

Data Gathering Instruments

The researchers used different techniques to gather the necessary data for the development and evaluation of the system.

A structured questionnaire was distributed to respondents to evaluate the system's usability, efficiency, and reliability. The questionnaire utilized a Likert scale to measure the level of agreement of respondents.

The collected responses helped determine whether the system is effective and user-friendly for barangay health workers, midwives, and officials handling residents' health records.

Interviews were conducted with barangay health workers, nurses, midwives, and officials to understand the current challenges in managing health records. This helped identify system requirements and necessary features.

Data Analysis

The data gathered from the survey questionnaires were analyzed using **descriptive statistics**, particularly the **weighted mean**, to determine the level of agreement of the respondents.

A **five-point Likert scale** was used:

Scale	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

The weighted mean was computed using the formula:

Mean = (Total Score) ÷ (Number of Respondents)

The results will be interpreted using the following scale:

Mean Range	Interpretation
4.20 – 5.00	Very Good
3.40 – 4.19	Good
2.60 – 3.39	Moderate
1.80 – 2.59	Poor
1.00 – 1.79	Very Poor

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